
TASK 2: REQUIREMENT GATHERING.

DESIGN AND IMPLEMENTATION OF A CROWDSENSED DRIVE TEST MOBILE APPLICATION

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1. Introduction

Mobile communication networks form the backbone of modern digital services, enabling voice communication, mobile broadband, e-learning, mobile banking, and e-government applications. As these networks expand and user demand grows, ensuring adequate radio coverage and acceptable Quality of Service (QoS) becomes increasingly critical for mobile network operators (MNOs).

Traditionally, network coverage assessment has relied on physical drive testing, where specialised equipment mounted in vehicles collects radio frequency (RF) measurements along predefined routes. While accurate, this approach is expensive, labour-intensive, and difficult to scale — particularly in developing regions with challenging terrain, limited infrastructure, and constrained budgets.

Recent advances in smartphone sensing capabilities, mobile operating systems, and wireless APIs have enabled ordinary smartphones to function as Class-2 IoT devices, capable of collecting geo-referenced network performance data in real time. This project leverages this capability to design and implement a crowdsensed drive test mobile application that collects drive-test-like RF measurements from smartphones and transmits them to a backend analytics platform for coverage prediction and coverage hole detection.

This document covers Task 2 of the project, the requirement gathering phase, documenting the stakeholders identified, the techniques used to gather requirements, the data collected, how that data was cleaned and structured, and the user reluctance assessment conducted to ensure the application meets real user needs and overcomes adoption barriers.

2. Stakeholder Identification

Stakeholder identification is a critical first step in the requirement gathering process. A stakeholder is any individual, group, or organisation with an interest in or that may be affected by the system being developed. For the crowdsensed drive test application, stakeholders were identified across primary user groups, secondary beneficiaries, technical teams, and regulatory bodies.

2.1 Primary Users.

Primary users are those who directly interact with the system on a regular basis.

Role	Primary Responsibility	System Interaction
Drive Test Technicians	Operate the mobile collector app, initiate test sessions, validate data quality on the go	Mobile app - daily use in vehicles
Network Planning Engineers	Consume the backend dashboard to identify coverage holes, plan new sites, and optimise antenna parameters	Web dashboard -post-processing analysis
RF Optimisation Engineers	Analyse RSRP/RSRQ/SINR trends to tune network performance and resolve interference issues	Web dashboard - RF parameter analysis
Data Scientists	Build coverage prediction and interpolation models using crowdsensed measurement data	Backend API — model training and deployment

2.2 Secondary Stakeholders

Secondary stakeholders benefit from the system's outputs without directly operating it.

Role	Interest in the System	Type
Network Operations Centre (NOC) Teams	Use coverage alerts and hole detection reports to prioritise incident response and dispatching	Secondary beneficiary
Executives / Management	Track high-level KPIs such as coverage percentage, QoS benchmarks, and cost savings vs traditional drive testing	Decision maker
Data Privacy Advocates	Ensure the system complies with GDPR and applicable local data protection laws, since user location data is being crowdsensed	Regulatory oversight
End Subscribers / Consumers	Indirectly benefit from network improvements driven by coverage hole identification and resolution	Indirect beneficiary

Role	Interest in the System	Type
Infrastructure / Tower Partners	May be engaged based on coverage hole analysis findings that indicate new site requirements	Indirect stakeholder

3. Requirement Gathering Techniques

A multi-technique approach was adopted to ensure comprehensive requirement coverage. Since direct interviews with mobile network operators were not possible, the team adapted its elicitation strategy to combine secondary research, reverse engineering, contextual inquiry, and brainstorming.

3.1 Secondary Research

The primary elicitation technique used was secondary research — the systematic review of existing academic literature, industry documentation, and online educational resources. The following sources were consulted:

- ScienceDirect - academic paper on mobile crowdsensing for network measurement (doi:10.1016/j.jnca.2016.02.018), which provided theoretical grounding for the crowdsensing architecture and measurement parameter selection
- iTelcoTech RAN Engineer 4G and 5G learning path - provided detailed technical context on LTE/5G RF measurement parameters, cell identity structures, and drive test protocols
- GeeksForGeeks Software Development Process guide - informed the structured development methodology adopted for the project
- GeeksForGeeks Data Collection Methods - provided methodological framework for structuring the data gathering process
- Online drive test session recordings and walk-throughs - provided contextual observations of real professional drive tests in operation.

3.2 Contextual Inquiry

Contextual inquiry involved following real drive test sessions documented online to observe actual operational conditions. This technique was adapted from its traditional form (physical ride-along) due to the lack of direct MNO access. Observations included:

- Vehicle setup: air conditioning required in the rear to prevent equipment overheating; large roof structure to house multiple antennas with sufficient separation to prevent RF interference
 - Equipment configuration: 4 voice phones and 4 data cards per carrier, each for incoming and outgoing calls; a Single Board Computer (SBC) controls both units; all controlled by a laptop in the front
 - GPS antenna placement: positioned centrally on the roof to ensure accurate geo-referencing of every measurement
 - Session duration: approximately 90 minutes per run; full shutdown and restart required after any unresolved call drop to prevent data contamination
 - Voice testing: Harvard sentences (phonetically balanced test phrases) used to evaluate speech quality under varying signal conditions, handovers, and interference
 - Data testing: continuous upload and download cycle; low-speed areas mapped and referred to engineers for capacity or infrastructure decisions
 - Post-session: benchmarking performed so that similar issues encountered in future areas can be addressed with known solutions.
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4. Data Gathering.

Data gathering for this project took two forms: operational data (how drive tests are conducted in practice) and technical data (what measurements the application must collect from smartphones).

4.1 Drive Test Operational Data

Through contextual inquiry of online drive test sessions, the following operational requirements were identified:

4.1.1 Vehicle and Equipment Setup

A professional drive test vehicle requires:

- Air conditioning in the vehicle rear to regulate temperature for electronics
- A structural roof mount large enough to house multiple antennas with adequate physical separation (minimum 30cm between voice and data antennas to prevent inter-antenna interference)
- 4 voice phones and 4 data cards per carrier — 2 varieties each for incoming and outgoing call testing
- A Single Board Computer (SBC) to control both voice and data units simultaneously
- A control laptop running drive test software (e.g. TEMS Investigation) in the front of the vehicle
- Window-mounted data antennas hooked to data cards, positioned away from equipment to minimise near-field interference
- Roof-mounted voice antennas for call testing; GPS antenna placed centrally on roof for accurate positioning

4.1.2 Session Procedure

1. Configure all equipment and verify all connections before departing
2. Initiate drive test session on control laptop — begins continuous voice call cycling and data test cycling
3. Voice calls average 90 seconds using Harvard sentences; calls continuous throughout the session
4. Data module performs upload and download tests continuously, stopping and repeating
5. If a call drops: monitor briefly for self-correction; if unresolved, stop the vehicle, shut down all equipment, and restart the session from the beginning
6. Areas with unsatisfactory data speeds are flagged and mapped for post-processing
7. Post-session benchmarking performed and findings passed to network engineers

4.1.3 Coverage Analysis Workflow

When coverage gaps are identified during post-processing, engineers determine the appropriate response. If capacity is insufficient, additional cards or radio equipment may be installed on existing towers. If there is no existing infrastructure in range, a new site may be recommended. In cases of high-

rise buildings with no indoor coverage, specialist in-building engineers (IBE) handle deployment of Distributed Antenna Systems (DAS).

4.2 RF Measurement Data Specification

Based on secondary research and contextual inquiry, the following measurement categories and specific parameters were identified as required data points for the application:

4.2.1 Radio / RF Measurements

Parameter	Unit	Description	RAT
RSRP	dBm	Reference Signal Received Power — primary LTE/5G coverage indicator	4G / 5G
RSRQ	dB	Reference Signal Received Quality — signal quality relative to interference	4G / 5G
RSSI	dBm	Received Signal Strength Indicator — total received wideband power	2G / 3G / 4G
SINR	dB	Signal to Interference + Noise Ratio — channel quality indicator	4G / 5G
CQI	0–15	Channel Quality Indicator — reported to base station for scheduling	4G / 5G
RSCP	dBm	Received Signal Code Power — 3G WCDMA equivalent of RSRP	3G
Ec/No	dB	Chip Energy to Noise Ratio — 3G signal quality indicator	3G
RxLev / RxQual	dBm / 0–7	2G GSM received level and quality metrics	2G
SS-RSRP / SS-SINR	dBm / dB	5G NR Synchronisation Signal Block measurements	5G NR

4.2.2 Cell Identity and Network Parameters

Parameter	Description
Physical Cell ID (PCI)	Serving cell identifier — maps measurement to a specific base station
EARFCN / NR-ARFCN	Absolute radio frequency channel number — identifies the carrier frequency
eNodeB / gNodeB ID	Base station identifier for LTE and 5G NR respectively
MCC / MNC	Mobile country and network codes — identifies the serving operator
LAC / TAC	Location / Tracking Area Code — coarse network topology context

Parameter	Description
Network type (RAT)	Active radio access technology: 2G GSM / 3G UMTS / 4G LTE / 5G NR
Frequency band	Active LTE or NR band — e.g. Band 3 (1800 MHz), n78 (3.5 GHz)

4.2.3 Location and Mobility Data

Parameter	Unit	Description
GPS latitude / longitude	Decimal degrees	WGS84 coordinates — primary spatial reference for all samples
GPS altitude	Metres	Height above ellipsoid — relevant for terrain-based coverage modelling
GPS accuracy	Metres	Horizontal position uncertainty — used to filter low-quality fixes
Speed over ground	km/h	Device speed — context for Doppler effects and handover analysis
Bearing / Heading	Degrees	Direction of travel — used to reconstruct drive route
UTC Timestamp	ms epoch	ISO 8601 UTC timestamp — synchronises samples across devices

4.2.4 Throughput and QoS Metrics

Parameter	Unit	Description
Downlink throughput	Mbps	Measured via active speed test or passive monitoring
Uplink throughput	Mbps	Important for upload-intensive applications
Round-trip latency (RTT)	ms	User-perceived responsiveness to a reference server
Packet loss	%	Percentage of lost packets during a test burst
Jitter	ms	Packet delay variation — critical for voice and video QoS evaluation

4.2.5 Device and Session Metadata

Parameter	Description
Device model	Manufacturer and model — controls for device-specific RF performance variation
OS version	Android / iOS version — affects API availability and measurement accuracy
Session ID	Unique drive test session UUID — groups all samples from a single run
User / Device ID	Anonymised contributor identifier — supports multi-device crowdsensing

Parameter	Description
Battery level	Device battery % — flag low-battery samples that may affect quality
Sample interval	Configured measurement polling frequency — target 1 sample per second

5. Data Cleaning

The raw data collected through secondary research, contextual inquiry, and user surveys contained noise, inconsistencies, and overlapping information that needed to be structured before it could inform requirements. The following six-step cleaning process was applied.

5.1 Step 1: Raw Data Capture and Consolidation

All raw data sources were consolidated into a single repository before cleaning:

- Timestamped field notes from contextual inquiry sessions
- Screenshots and annotations from competitive analysis of TEMS Pocket, G-NetTrack, NetMonster, and OpenSignal
- Annotated academic paper excerpts and course material highlights
- Raw user reluctance survey responses
- Team brainstorming session notes

5.2 Step 2: Noise Removal

Irrelevant or non-generalisable data was removed from the dataset. Filtered items included:

- One-off personal habits unlikely to represent the broader user group (e.g. a single user's preference for a specific screen layout not shared by others)
- Tool limitations specific to competitor technology stacks that would not apply to our architecture (e.g. TEMS Pocket's hardware dongle requirements)
- Environmental observations about non-app factors such as road conditions, weather, and vehicle performance
- Duplicate observations captured from multiple sources that described the same finding

5.3 Step 3: Pattern Identification

The data was analysed for convergent patterns (strong design signals) and contradictions (design decisions requiring resolution):

5.4 Step 4: Synthesis into Design Artefacts

The cleaned and structured data was synthesised into the following actionable design artefacts used as inputs to the requirement analysis phase:

- User personas: Drive Test Technician, RF Planning Engineer, NOC Analyst
 - User journey maps: session setup, active data collection, call drop response, post-session upload
 - UI convention inventory: metric positions, colour coding standards, control placement patterns
 - Prioritised pain point list: fed directly into the requirement backlog
 - 'How might we' statements: e.g. 'How might we ensure the app is fully operable with one hand while in a moving vehicle?'
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6. User Reluctance Assessment

User reluctance assessment involves identifying, measuring, and understanding resistance that potential users have towards adopting a new system. This is particularly important for this application because it asks users to change established workflows, trust a smartphone as a measurement device, and share location data with a backend platform. Reluctance was assessed through informal surveys and structured analysis using the Technology Acceptance Model (TAM) framework.

6.1 Reluctance Triggers Identified

The following application-specific reluctance triggers were identified before data collection began, based on the nature of the system:

Trigger	Why It Causes Reluctance	Risk Level
Smartphone as measurement device	Scepticism about accuracy vs dedicated calibrated hardware	High
Crowdsensed / shared data model	Privacy concerns; questions about data ownership and who sees the data	High
Always-on data collection	Battery drain anxiety; mobile data quota concerns	Medium
Backend platform they don't control	Trust questions about data storage security and handling	High

6.2 Survey Responses Collected

Informal surveys conducted with target users produced the following direct responses, which were analysed and categorised:

6.2.1 Statements of Concern

- **"I don't want my location tracked"**

Category: Privacy. Frequency: High. This is the most commonly expressed concern. Users are uncomfortable with continuous GPS logging even when they understand it is necessary for the app's core function.

- **"I don't want it taking too much space in my phone"**

Category: Storage. Frequency: Medium. Users are concerned about app installation size and local data buffer accumulation during long sessions without connectivity.

- **"I don't want my information exposed"**
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Category: Data security. Frequency: High. Users want assurance that their measurement data — particularly location traces — cannot be accessed by unauthorised parties or used for surveillance purposes.

6.2.2 Questions Raised

- **"Can the app operate offline?"**

This question was raised by every field-oriented respondent. It directly maps to a critical functional requirement: the app must buffer data locally during connectivity gaps and upload automatically upon reconnection.

- **"Would the data be trusted?"**

This question reflects professional scepticism about using consumer smartphone hardware in place of calibrated drive test equipment. It requires the application to include a validation or benchmarking feature that demonstrates measurement accuracy comparable to professional tools.

- **"Would the app be easy to use?"**

This question reflects concern about the learning curve, particularly from users familiar with existing tools. It reinforces the need for a UI that mirrors familiar conventions from TEMS Pocket and similar tools.

- **"How long will it take?"**

This question has three interpretations: session duration, data upload time, and time to see results on the backend dashboard. Each interpretation must be addressed with specific SLAs defined in the non-functional requirements.

6.3 Reluctance Risk Matrix

Concern	Frequency	Impact	Risk	Design Response
Privacy / location tracking	High	High	Critical	Anonymise device IDs before transmission; explicit consent flow; GDPR-compliant privacy policy
Data security / exposure	High	High	Critical	TLS 1.3 encryption in transit; AES-256 at rest for local buffer
Smartphone accuracy not trusted	High	High	Critical	Side-by-side TEMS Pocket benchmark shown in onboarding; confidence interval displayed with readings
Offline operation	High	High	Critical	Full offline mode; 90-minute local buffer; auto-upload on reconnection with progress indicator
Storage consumption	Medium	Medium	Moderate	App target ≤50 MB; local buffer auto-purged after confirmed upload

Concern	Frequency	Impact	Risk	Design Response
Ease of use	Medium	High	High	Familiar TEMS Pocket UI conventions; one-handed design; contextual help tooltips
Battery drain	Medium	Medium	Moderate	Power-save mode with reduced sampling rate; battery level displayed; warning below 20%
Workflow disruption	Low	Medium	Low	Parallel mode allows simultaneous use alongside existing tools during transition period

7. Summary and Coverage Assessment

The requirement gathering phase successfully collected data across all five required dimensions. The table below summarises coverage and identifies items that require follow-up in the requirement analysis phase.

Section	Status	Key Gaps to Address in Task 3
Stakeholder identification	Complete	None, all 8 roles documented with responsibilities
Requirement gathering techniques	Complete	MNO interviews not achieved, addressed via secondary research
Data gathering	Complete	5G NR scope needs explicit confirmation; L3 message feasibility needs resolution
Data cleaning	Complete	Affinity map to be validated with RF engineer stakeholder
User reluctance assessment	Complete	TAM survey should be repeated with larger sample before launch

7.1 Key Findings

- The most critical user concern is privacy — location tracking and data exposure. All design decisions must treat these as non-negotiable constraints rather than optional features.
- Offline operation is not a nice-to-have — it is a fundamental requirement given the field context. Sessions last up to 90 minutes in areas that may have no data connectivity.
- Trust in smartphone measurement accuracy is the single biggest barrier to professional adoption. A benchmarking/validation feature must be included in the first release.
- UI design must draw heavily on TEMS Pocket conventions — professional users will resist an application that forces them to relearn muscle memory developed over years of drive testing.
- Data gathering revealed a scope conflict between the crowdsensed smartphone model and the professional vehicle rig model. This conflict must be resolved in Task 3 before requirements can be finalised.

7.2 References

- iTelcoTech. RAN Engineer 4G and 5G Learning Path.
<https://www.itelcotech.com/learningpath/ran-engineer-4g-and-5g>
 - GeeksForGeeks. Software Development Process — Step-by-Step Guide.
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